

Python 資料分析與機器學習 期末專題模板

Final Project Paper Template for Data Analysis and Machine Learning with Python

(整篇文章統一用中文或英文寫)

(The entire article must be written consistently in either Chinese or English.)

Title (標題)

描述系統支援的決策任務與應用情境 (不要只寫技術名稱)

Describe the decision-making tasks that the system supports and the practical scenarios in which it can be applied. (Do not simply list the names of technologies.)

Authors (作者)

Name 1, Name 2, Name 3, ...

(未來投稿的作者順序)

(The order of authors for future submissions)

Abstract (摘要) (200~250 words)

請以四個明確段落或句群撰寫，務必涵蓋：

Write in four clearly defined paragraphs or sentence groups, and make sure to cover the following:

1. 系統使用情境與決策背景

System Usage Scenarios and Decision-Making Context

(Application Context)

2. 設計目標

Design Objective

3. 主要設計取捨 (至少2~3 個)

Key Design Choices and Trade-offs (at least 2~3)

4. 系統如何被評估 (非使用者實驗)

Explain how the system is evaluated (excluding user studies)

I. 應用背景和決策問題

Application Context and Decision Problem

內容應包含：

The content should include:

- 系統應用情境
System Application Scenarios
- 使用者面臨的決策困難
Decision-Making Challenges Faced by Users
- 為何現有做法不足
Limitations of Existing Approaches
- 本系統欲支援的具體決策或判斷
Specific Decisions or Judgments Supported by the System

II. 相關設計方法

Related Design Approaches

內容應包含：

The content should include:

- 相關系統或既有解法「如何設計」
Design of Related Systems or Existing Solutions
- 它們在此情境下的限制
Limitations of These Systems in This Context
- 本系統與其設計取向的差異
Differences Between This System and Their Design Approaches

III. 系統概述

Artifact Overview

內容應包含：

The content should include:

- 系統整體功能
Overall System Functions
- 系統邊界 (做什麼 / 不做什麼)
System Boundaries (What It Does and Does Not Do)
- 使用流程 (建議附系統流程圖或架構圖)
Usage Workflow (It Is Recommended to Include a System Flowchart or Architecture Diagram)

IV. 設計決策與權衡

Design Decisions and Trade-offs

請至少包含 3 個設計決策小節，例如：

Please include at least three subsections on design decisions, for example:

Decision 1: Interpretability vs. Predictive Accuracy

Decision 2: Automation vs. Human Control

Decision 3: Model Complexity vs. Maintainability

註：每一小節必須說清楚：

Note: Each subsection must clearly explain the following:

- 為何選擇此方案
Why this option was chosen
- 放棄了哪些替代方案
Which alternative options were rejected
- 適用條件與限制
The conditions and limitations under which it applies

V. 人工智慧角色和系統自主性

AI Role and System Autonomy

內容應包含：

The content should include:

- AI 在系統中的角色 (預測、建議、分析、模擬)
The Role of AI in the System (Prediction, Recommendation, Analysis, Simulation)
- 系統自動化程度
Level of System Automation

- 使用者如何介入、理解或覆核 AI 輸出
How Users Intervene, Understand, or Review AI Outputs

VI. 評估系統、邏輯和證據

Evaluation System, Logic and Evidence

內容應包含：

The content should include:

- 評估目標與方法 (同時判斷此設計是否合理)
Evaluation Goals and Methods (Including Assessment of Design Appropriateness)
- 評估的情境
Evaluation Scenarios
- 比對的基準線 (規則法、簡化模型、人工判斷)
Comparison Baselines (Rule-Based Methods, Simplified Models, Human Judgments)
- 效能與結果 (可為真實系統測試或模擬測試)
Performance and Results (From Real System Testing or Simulation)
- 失敗案例或結果不如預期的案例
Failure Cases or Cases with Unexpected Outcomes
- 評估方法限制
Evaluation Methods' Limitations

VII. 討論：可遷移的設計概念

Discussion: Transferable Design Insights

(說明：從你的系統設計或實驗經驗中，抽取出的一般性原則、策略或教訓，這些概念不僅適用於你的特定系統，也可以應用

到其他類似場景或系統中。

Explanation: General principles, strategies, or lessons drawn from your system design or experimental experience. These concepts are not only applicable to your specific system but can also be applied to other similar scenarios or systems.)

內容應包含：

The content should include:

- 哪些設計具有可轉移性？
Which Design Elements Are Transferable?
- 哪些高度依賴情境？
Which Are Highly Context-Dependent?
- 設計風險與治理考量
Design Risks and Governance Considerations

VIII. 結論與未來展望

Conclusion and Future Extensions

內容應包含：

The content should include:

- 總結系統設計貢獻
Summary of System Design Contributions
- 若資源增加，設計上會如何調整
How the Design Would Be Adjusted if More Resources Were Available

參考文獻

References

格式統一 (APA / IEEE / ACM 皆可)

Consistent formatting (APA, IEEE, or ACM style is acceptable)

範例 (僅作為文章結構和內容說明, 並非真實例子)

標題

一個可解釋人工智慧的課程學習風險與退選決策支援系統

作者

張三、李四、王五

摘要

在大學學習過程中, 學生經常需要在學期中期判斷是否應該繼續修習某一門課程, 尤其是在成績表現不穩定或時間壓力增加的情況下。傳統的成績查詢系統通常僅提供目前分數與簡單加權計算, 缺乏對未來表現的不確定性分析, 也難以協助學生理解影響成績的重要因素。因此, 本研究提出一個可解釋人工智慧決策支援系統, 用於協助學生在課程學習過程中評估風險並探索不同學習情境下的可能結果。

本系統的設計目標在於提供一個透明且可理解的分析工具, 使學生能夠透過情境模擬了解不同努力程度、作業表現與考試結果可能帶來的成績範圍。系統整合歷史課程資料、成績權重規則以及可解釋機器學習模型, 產生多種可能結果與風險解釋, 而非僅提供單一預測值。

在系統設計過程中, 本研究面臨多項重要設計取捨。首先, 在模型選擇上優先考量可解釋性而非最高預測準確度, 以確保學生能理解分析結果。其次, 在系統角色上採用「建議式決策支援」而非自動決策, 以維持使用者自主性。此外, 系統採用情境模擬而非單點預測, 使結果更符合實際學習環境中的不確定性。

系統的評估並未採用使用者實驗, 而是透過情境模擬分析、與規則式成績計算工具進行比較, 以及分析預測失敗案例等方式進行驗

證。評估結果顯示，本系統能在多數情境下提供合理且具解釋性的分析結果，並能協助學生更全面地理解學習風險與可能的改善策略。

I. 應用背景和決策問題

在大學學習環境中，學生在學期中期常常需要評估自己的學習狀況並決定是否繼續修習某些課程。當課程難度較高或成績表現不理想時，學生必須在有限資訊下判斷是否繼續投入時間與精力，或選擇退選以避免影響整體學業成績。這種決策通常涉及多種不確定因素，例如未來考試成績、作業表現、時間投入程度以及課程評分規則等。

然而，目前大多數學校的教務系統僅提供靜態的成績查詢與簡單的加權計算功能。學生雖然可以看到目前的分數，但很難理解在不同努力程度或不同考試表現下可能出現的最終成績範圍。此外，現有工具通常無法說明哪些因素對最終成績影響最大，因此學生往往只能依賴直覺或與同儕討論來做決策。

本研究所提出的系統旨在支援一個具體的決策問題：在學期進行到某一時間點時，學生是否應該繼續修習某門課程。系統並不直接替學生做出決定，而是透過資料分析與情境模擬，呈現不同努力程度與表現情境下可能出現的成績範圍，並解釋哪些因素對結果影響最大。透過這種方式，學生能夠在更充分的資訊基礎上進行判斷。

II. 相關設計方法

過去針對學習表現分析的系統通常採用兩種主要方法。第一類方法是基於規則的成績計算工具，這類系統根據課程的評分比例計算最終分數，具有高度透明性，但無法處理未來表現的不確定性。第二類方法則是利用機器學習模型預測學生最終成績，例如使用回歸模型或神經網路進行預測。雖然這些方法可以提高預測準確度，但其結果通常缺乏可解釋性，使學生難以理解預測背後的原因。

在本研究的應用情境中，單純追求預測準確度並非最重要的目標。學生更需要的是理解不同學習策略可能帶來的結果，以及哪些因素最值得優先改善。因此，本系統採取不同的設計取向，將焦點放在決策支援而非純粹預測。

與既有方法相比，本系統的設計強調三個特點：情境模擬、多結果呈現以及可解釋分析。透過模擬不同努力程度與作業表現情境，系統可以提供多種可能結果，而非單一預測值。同時，系統利用可解釋模型分析各項因素的重要性，使學生能理解影響成績的關鍵因素。

III. 系統概述

本研究所提出的系統是一個基於 Python 的分析型決策支援系統。使用者首先輸入目前的課程資料，包括作業成績、小考分數、出席率以及課程評分比例等資訊。系統接著整合歷史資料與預測模型，生成多種可能的最終成績情境。

系統的主要輸出包含三個部分：第一，根據不同未來表現情境估計可能出現的成績區間；第二，分析影響成績的重要因素並提供解釋；第三，提供可能的改善建議，例如提高作業完成率或增加考試準備時間。

在系統邊界方面，本系統僅處理與課程學習相關的資料，不收集學生的個人敏感資訊，也不會自動做出任何課程決策。其角色僅為分析與建議工具，最終決策仍由使用者自行判斷。

系統使用流程大致包含三個階段。首先，使用者輸入目前成績與課程規則。其次，系統進行資料分析與情境模擬。最後，系統以圖表與文字說明方式呈現結果，使使用者能理解不同情境下的可能 outcomes。

IV. 設計決策與權衡

Decision 1: 情境模擬 vs. 單點預測

在系統設計過程中，一個重要決策是採用情境模擬而非單一預測值。雖然單點預測能提供看似精確的結果，但在實際學習環境中，未來表現具有高度不確定性。情境模擬可以呈現多種可能結果，使學生能更全面地理解風險。然而，這種方法也可能增加系統複雜度，因此需要透過視覺化方式降低理解難度。

Decision 2: 可解釋性 vs. 預測準確度

本系統選擇使用線性回歸與決策樹等可解釋模型，而非深度學習模型。這種設計可能在某些情境下降低預測準確度，但能讓使用者清楚了解哪些因素對結果影響最大，符合決策支援系統的設計目標。

Decision 3: 建議式決策支援 vs. 自動決策

系統並不直接給出「應該退選」或「應該繼續」的建議，而是提供分析結果供學生參考。這種設計可以避免過度依賴系統，也降低倫理與責任問題。然而，這也意味著使用者需要自行解讀結果，因此系統必須提供清楚的解釋與視覺化資訊。

V. 人工智慧角色和系統自主性

在本系統中，人工智慧主要扮演分析與預測的角色。AI 模型負責分析歷史資料與目前輸入的成績資訊，估計可能的成績範圍並識別重要影響因素。系統同時利用解釋方法顯示各項因素對結果的影響程度。

系統的自動化程度相對有限。AI 僅提供分析結果與建議，不會自動做出任何決策或改變學生的課程狀態。使用者可以查看分析結果、調整輸入參數，並重新生成不同情境的模擬結果。透過這種方式，使用者能夠理解 AI 的推論過程並保有最終決策權。

VI. 評估系統、邏輯和證據

本系統的評估主要透過三種方法進行。首先，我們將系統輸出與傳統規則式成績計算工具進行比較，以確保在標準情境下結果一致。其次，我們透過情境模擬測試系統在不同學習狀況下的表現，例如高作業分數但低考試分數的情境，或出席率不穩定的情境。

此外，我們也分析了一些預測失敗案例。例如，在極端成績分布或歷史資料不足的情況下，模型可能產生較不穩定的預測結果。透過分析這些案例，我們能夠理解系統的限制並提出改進方向。

評估結果顯示，系統在多數情境下能提供合理的分析與解釋，但在資料量較少或課程評分規則變動較大的情況下，其預測能力可能受到影響。

VII. 討論：可遷移的設計概念

從本研究的系統設計與實驗過程中，可以歸納出一些具有可轉移性的設計原則。例如，在不確定性較高的決策情境中，情境模擬往往比單點預測更能幫助使用者理解風險。此外，可解釋模型在決策支援系統中通常比高準確度但不透明的模型更具實用價值。

然而，本系統的一些設計也高度依賴特定情境。例如，系統假設課程評分規則相對穩定且可量化，因此在評分方式高度主觀或不透明的課程中，其適用性可能降低。

在設計風險與治理方面，系統必須避免被誤用為「自動決策工具」。若學生過度依賴系統而忽略其他因素，可能導致不理想的決策。因此，在系統介面設計中需要清楚強調其建議性質。

VIII. 結論與未來展望

本研究提出一個可解釋人工智慧決策支援系統，用於協助學生在學期中期評估課程學習風險並做出是否繼續修課的決策。系統透過情境模擬與可解釋模型提供多種可能結果與風險分析，並強調使用者理解與自主決策的重要性。

未來若有更多資源與資料，本系統可以進一步整合更多學習行為資料，例如學習時間、作業提交紀錄與線上學習平台活動。此外，也可以發展更進階的互動視覺化工具，使學生能更直觀地探索不同學習策略可能帶來的結果。

參考文獻

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Gregor, S., & Hevner, A. (2013). Positioning and presenting design science research for maximum impact. *MIS Quarterly*, 37(2), 337–355.

Molnar, C. (2022). *Interpretable Machine Learning*. Lulu Press.

Shmueli, G., Bruce, P., Yahav, I., Patel, N., & Lichtendahl, K. (2018). *Data Mining for Business Analytics*. Wiley.

Example (for illustrating article structure and content only; not a real case)

Title

An Explainable AI Decision Support System for Course Continuation and Grade Risk Assessment

Authors

Student A, Student B, Student C

Abstract

University students frequently encounter difficult decisions during the middle of a semester when determining whether they should continue taking a course or withdraw from it. This decision is often complicated by incomplete information, such as partial grades, uncertain future performance, and limited understanding of how different course components influence final outcomes. Existing academic information systems typically provide static grade records and simple score calculations, offering little support for understanding academic risks or exploring alternative learning scenarios. As a result, students often rely on intuition or peer advice when making important academic decisions.

The objective of this project is to design an explainable AI-enabled decision support system that helps students evaluate their academic standing and explore potential outcomes under different scenarios. The system integrates course grading rules, historical academic data, and

interpretable machine learning models to generate grade range estimations and identify key factors influencing performance. By presenting scenario-based outcomes rather than a single prediction, the system encourages users to understand the uncertainty inherent in academic performance.

Several key design trade-offs were considered in the development of the system. First, interpretability was prioritized over maximum predictive accuracy to ensure that users can understand the reasoning behind system outputs. Second, the system adopts an advisory role rather than automated decision-making, preserving user autonomy. Third, scenario-based simulations are used instead of single-point predictions in order to better represent real-world uncertainty.

The artifact is evaluated using scenario-based simulations, comparisons with rule-based grade calculators, and the analysis of failure cases. The evaluation results indicate that the system can effectively support academic decision-making while maintaining transparency and user control.

I. Application Context and Decision Problem

During an academic semester, university students often face the challenging task of evaluating their learning progress and deciding whether to continue or withdraw from certain courses. This decision becomes particularly difficult when academic performance is uncertain or when students are struggling to keep up with coursework. Factors such as incomplete assignments, unexpected exam results, and time constraints can all contribute to uncertainty about final outcomes.

In most universities, academic information systems provide only basic grade reporting functions. Students can view their current scores or calculate estimated final grades based on fixed grading rules. However, these tools typically assume

deterministic outcomes and do not account for uncertainty in future performance. Furthermore, they rarely provide explanations about which factors are most influential in determining the final grade.

The decision problem addressed in this study is how students can determine whether continuing a course is a reasonable choice given their current academic status and potential future performance. Instead of generating a single predicted grade, the proposed system aims to support students by presenting multiple possible outcomes and explaining the factors that most strongly influence these outcomes. This approach allows students to better understand the risks and opportunities associated with different learning strategies.

II. Related Design Approaches

Existing systems for analyzing student academic performance generally follow two main design approaches. The first approach relies on rule-based grade calculators that use predefined grading weights to compute final scores. These systems are transparent and easy to understand, but they cannot capture uncertainty in future performance or provide deeper insights into performance risks.

The second approach involves the use of predictive machine learning models that estimate final grades based on historical student data. While these models may achieve higher predictive accuracy, they often function as black boxes, providing limited explanations for their predictions. As a result, users may find it difficult to trust or interpret the outcomes generated by these systems.

The design proposed in this study differs from both approaches by emphasizing decision support rather than prediction accuracy. Instead of generating a single predicted grade, the

system focuses on scenario-based analysis and interpretable explanations. By allowing users to explore multiple possible outcomes and understand the underlying factors influencing these outcomes, the system provides a more meaningful form of decision support.

III. Artifact Overview

The proposed artifact is an AI-enabled analytical decision support system implemented using Python-based interface. Users interact with the system by entering current course-related information, including assignment scores, quiz results, attendance records, and the course grading structure. Based on these inputs, the system analyzes the data and generates simulated scenarios of possible final grade outcomes.

The system produces three primary outputs. First, it estimates possible grade ranges under different future performance scenarios. Second, it identifies key factors that influence the predicted outcomes and provides explanations for these influences. Third, it generates actionable suggestions that may help students improve their academic performance.

The system boundary is intentionally limited. It only processes course-related performance data and does not collect sensitive personal information. Moreover, the system does not make enrollment decisions on behalf of students. Its role is strictly advisory, supporting human judgment rather than replacing it.

The typical workflow begins with user data input, followed by system analysis and scenario simulation. The results are then presented through visualizations and textual explanations that help users understand the potential consequences of different academic choices.

IV. Design Decisions and Trade-offs

Decision 1: Scenario-Based Simulation vs. Single-Point Prediction

One key design decision was to present grade outcomes as ranges generated from multiple simulated scenarios rather than as a single predicted value. Single-point predictions may appear precise but can be misleading when uncertainty is high. Scenario-based simulation allows users to explore different possible futures and better understand the range of potential outcomes. However, this approach requires careful visualization and explanation to avoid overwhelming users.

Decision 2: Interpretability vs. Predictive Accuracy

The system prioritizes interpretable models such as linear regression and decision trees instead of complex deep learning models. While this choice may reduce predictive accuracy in some cases, it allows the system to explain which variables most strongly influence grade outcomes. This transparency aligns with the goal of supporting informed decision-making.

Decision 3: Advisory Support vs. Automated Decision-Making

Another design decision was to position the system as an advisory tool rather than an automated decision-making system. Instead of recommending whether a student should withdraw from a course, the system provides analytical insights that users can consider when making their own decisions. This design preserves user autonomy while reducing ethical and governance concerns.

V. AI Role and System Autonomy

In this system, artificial intelligence primarily functions as an analytical advisor. The AI component analyzes historical data

and user-provided inputs to estimate potential grade outcomes and identify influential factors affecting academic performance.

The level of automation in the system is intentionally limited. AI-generated outputs are presented as analytical insights and scenario simulations rather than deterministic decisions. Users are encouraged to review the explanations, explore alternative scenarios, and interpret the results before making their own decisions. In this way, the system supports human decision-making while maintaining transparency and user control.

VI. Evaluation System, Logic and Evidence

The system is evaluated using three complementary approaches. First, its outputs are compared with those generated by traditional rule-based grade calculators to ensure consistency under standard grading scenarios. This comparison verifies that the system behaves logically when deterministic inputs are provided.

Second, scenario-based simulations are used to test the system across a wide range of academic performance situations, such as high assignment scores combined with low exam performance or irregular attendance patterns. This analysis helps determine whether the system provides meaningful insights across diverse conditions.

Third, failure cases are examined to identify situations where predictions become unstable or less reliable. For example, when historical data are limited or when grading rules change significantly, the model may produce less accurate estimates. Analyzing these cases helps clarify the limitations of the system.

The evaluation results suggest that the system can provide reasonable analytical insights in most situations, although its reliability may decrease in cases with insufficient data or highly variable grading policies.

VII. Discussion: Transferable Design Insights

Several design principles derived from this project may be transferable to other decision support contexts. In environments characterized by high uncertainty, scenario-based simulations can often provide more useful insights than single-point predictions. Additionally, interpretable models may offer greater practical value than highly complex models when users need to understand the reasoning behind system outputs.

However, some aspects of the system are highly context-dependent. For instance, the effectiveness of the system relies on the availability of structured grading rules and quantifiable performance metrics. In environments where evaluation criteria are highly subjective or inconsistent, the system may be less applicable.

From a governance perspective, it is important to ensure that users understand the advisory nature of the system. Overreliance on automated analyses could lead to suboptimal decisions if users ignore contextual factors not captured by the system.

VIII. Conclusion and Future Extensions

This study presents an explainable AI decision support system designed to assist university students in evaluating academic performance risks and making informed decisions about course

continuation. By combining scenario simulation with interpretable machine learning models, the system provides analytical insights that help users understand potential outcomes and influencing factors.

Future extensions of the system may include integrating additional learning behavior data, such as study time, assignment submission patterns, or online learning platform activity. With additional resources, more advanced visualization tools could also be developed to enhance users' ability to explore different academic scenarios interactively.

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